GIS for ICT Procurement by Government Agencies

Team 3.14159

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CS 191 WFQ1

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I. Customer Statement of Requirements

The Problem

Problem Description

Every year, Philippine government offices (city halls, schools, hospitals, provincial offices, national agencies) buy computers, laptops, printers, routers, software licenses, CCTV systems, and other ICT equipment and services. Because this is public money, the law requires these purchases to be announced and awarded openly. Most of this information ends up on **PhilGEPS (the Philippine Government Electronic Procurement System)**, the government's official procurement website.

The information is public, but it is not usable. PhilGEPS is confusing to navigate, and it offers no way to compare purchases across agencies, places, or years. Nobody can easily answer a simple question like "How much did government agencies pay for a laptop in Manila compared with Cebu?"

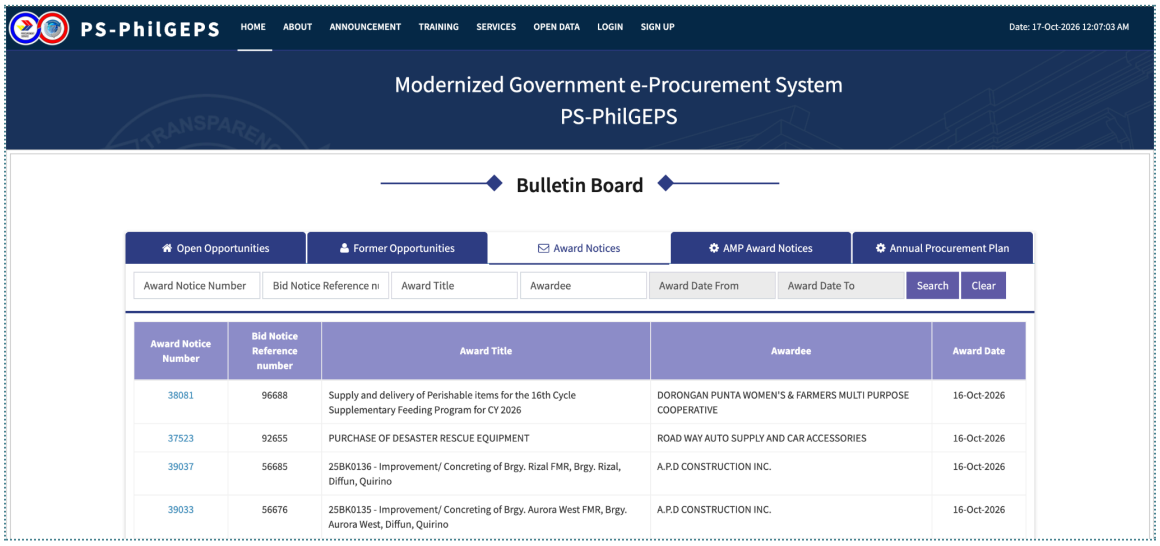
People Affected

Who	What they currently struggle with
Procurement officers in government agencies	They must set a budget for an ICT purchase but cannot easily see what other agencies actually paid for the same item.
Auditors	They suspect overpricing but must dig through records one at a time to prove it.
Suppliers and bidders	They want to price competitively but have no overview of what is being paid and where.
Researchers and students	They want to study government spending but must first collect and clean the data by hand.
Ordinary citizens	They want to know whether their taxes are spent wisely but cannot get a straight answer.

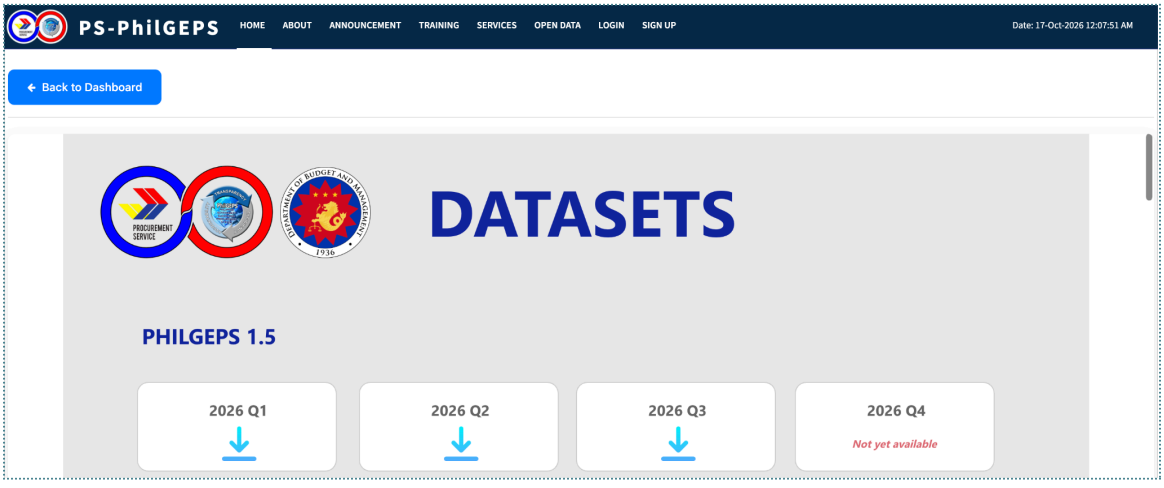
Current Handling

PhilGEPS gives users only two ways to access the data:

- A **search portal** that looks up notices by keyword or reference number. This only works if you already know what you are looking for. You get a list of notices, with no summary, no price comparison, and no way to see results by location. Also, not all award titles list the specific items provided, so some results might not show up.



- **CSV** files of the notices. These are raw dumps of records, not organized for comparing prices. Item descriptions are free text, so the files are hard to read, filter, and combine. In addition, these files include all types of bids (closed, awarded, active, cancelled, failed, and shortlisted), meaning that to get the actual finished projects, you still need to filter the datasets further manually.



Neither option lets a user ask “what did agencies pay for this item, and where?” So anyone who wants to compare ICT prices must do the following by hand:

- Search the portal for a keyword (for example, “laptop”) or download the CSVs.
- Open each notice one at a time to find the agency, budget, date, and award details.
- Copy the details into a spreadsheet.
- Repeat for dozens or hundreds of notices.
- Judge whether “Laptop, i5, 8GB” in one notice is comparable to “Notebook computer Core i5 8GB RAM 256GB SSD” in another.
- Work out where each agency is located, since the records list agency names, not map locations.
- Check which projects were actually completed, since cancelled, failed, and ongoing ones appear alongside them.

This is **slow, tedious, and error-prone**, so most people simply don’t do it.

Project Motivation

Without an easy way to compare, agencies fall back on supplier quotations, which may be inflated, or on last year’s price. Overpricing, when it happens, is hard to see. Comparing prices across places would help:

- Agencies to set fair budgets before they buy.
- The public to look for unusual prices.
- Suppliers to understand the market.
- Researchers to study government ICT spending without weeks of manual data cleaning.

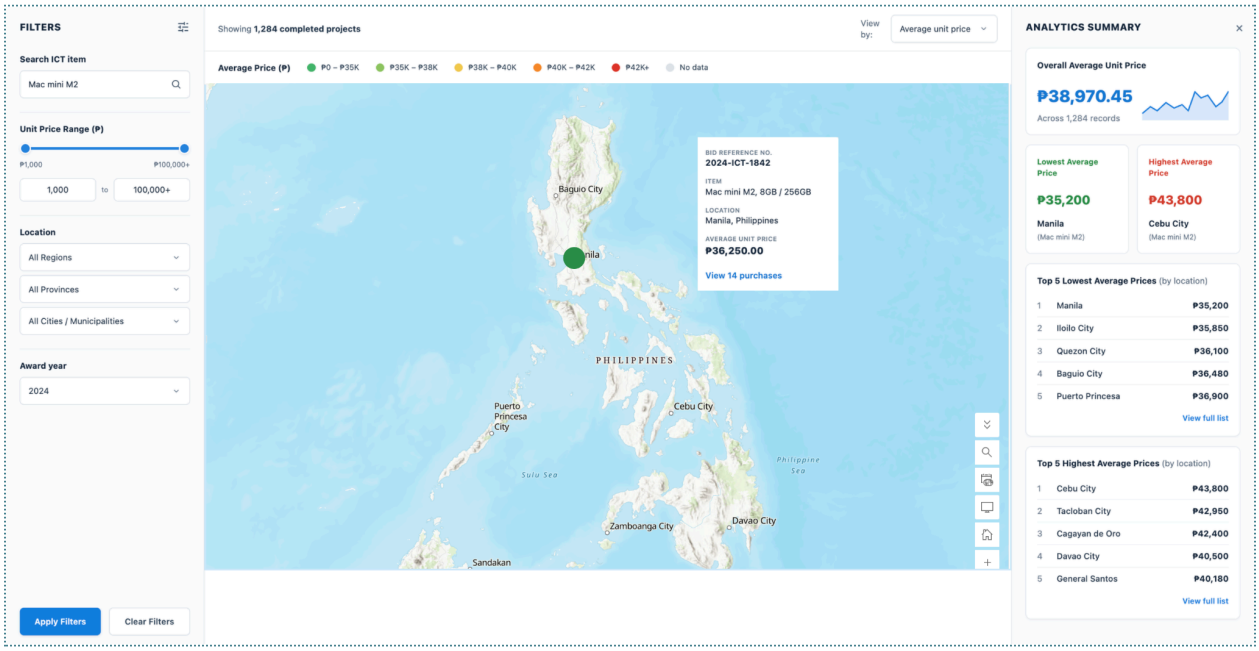
We aim to build a **Geographic Information System (GIS)** for ICT procurement. In simple terms, a GIS is a map connected to a database. Instead of digging through the website one notice at a time, a user sees prices on a map of the Philippines, clicks on certain agencies’ locations, and compares what was paid.

Key Requirements

- **Searching and Filtering**
    - The user shall be able to search for an ICT item by name or keyword (e.g., “Mac Mini”, “laptop”, “router”).
    - The user shall be able to filter results by location (region, province, city/municipality).
    - The user shall be able to filter results by year or date range.
    - The system shall show only completed ICT projects in all searches.
- **Map View**
    - The user shall be able to view the Philippines on an interactive map showing the prices of a selected item by location of a certain agency.
    - The system shall color-code locations on the map from lower to higher prices.
    - The user shall be able to zoom and pan the map.
    - The user shall be able to click a location and see a list of the purchases made there.
- **Price Comparison**
    - The user shall be able to compare the price of the same item between two or more locations (e.g., UP Manila vs UP Diliman).
    - The system shall show the average, lowest, and highest price of an item.
    - The system shall show the price difference between locations being compared.
    - The system shall highlight purchases whose prices are far above or below the typical price for the same item.
- **Purchase Details**
    - The user shall be able to open a single purchase record and see its details: agency, location, project title, budget, contract price, award date, supplier, and item list.
    - The system shall provide a link from each record to the original PhilGEPS notice so the user can verify it.
- **Access and Usability**
    - Anyone shall be able to use the map and charts without creating an account.
    - The system shall work in common web browsers on both desktop computers and mobile phones.
    - The system shall display the date when the data was last updated.

Preliminary UI Design

Note that the following is just a mockup and is not the final design / implementation for the project. These are just initial ideas of what the content of the page should look like.



The site has three main parts. On the left is a filter panel where you type the ICT item you want (like “Mac Mini”), then narrow the results by price range, location (region, province, or city), and award year, and click “Apply Filters.” In the middle is a map of the Philippines with colored dots for each location, where the color shows how expensive the item is there, from green (cheaper) to red (more expensive). Clicking a dot opens a small popup with the average price and a link to see the individual purchases, and from there you can open one purchase to see its full details and a link to the original PhilGEPS notice. On the right is a summary panel that shows the overall average price, the locations with the lowest and highest prices.

II. Plan of Work

a. Estimates of Cost and Resources

Human Resources

The team consists of us three student developers working alongside academic coursework across the 12-week project lifecycle (September 13, 2026 – December 5, 2026). In line with academic project constraints, we will not be budgeting monetary compensation but instead will be tracked via person-hours.

Human Resources			
Member	**Role**	**Responsibility**	**Est. Hours**
Jason Patrick Fangonil	*Full-Stack Developer/ Frontend Lead*	Leads the interactive map interface, purchase detail views, and front-end development	6-10hrs /week
Ethan Kurt Espejo	*Full-Stack Developer/ Systems and QA Lead*	Leads data ingestion, record validation, system architecture testing, and quality assurance	6-10hrs /week
Ambernie Salting	*Full-Stack Developer/ Backend Lead*	Leads database design, agency location mapping, and backend search filters.	6-10hrs /week
Total Human Resources		**~216 total hours**	

System Resources (Hardware & Software)

System resources encompass the hardware, runtime environment, data storage, and software tools that we will need to develop, host, and operate the system. The project makes use of existing client hardware and developer free tiers to keep costs at minimum.

System Resources			
Item	**Technical Requirement**	**Purpose**	**Est. Cost**
Database	*Standard SQL Database (e.g. MySQL)*	Storing procurement records, handling relational data, and structuring query schemas	Free
Web Hosting	*Vercel (free tier)*	Hosting the web application frontend and serving user interfaces	Free
GIS & Mapping	*QGIS / ArcGIS*	Spatial data processing, map creation, layer styling, and interactive web mapping	Free
Data Scraping & Processing	*Python (BeautifulSoup, Pandas)*	Ingesting, cleaning, transforming, and normalizing raw datasets	Free
Total System Cost		**₱0.00**	

b. Schedules and Milestones

For this project, we will assume two schedules: one for CS 191, and another for CS 192. For CS 191, the schedule starts with Week 1 (on the week of September 27, 2026), and ends on Week 9 (on the week of November 22, 2026). For CS 192, the schedule starts with Week 1 (on the week of January 27, 2027), and ends on Week 15 (on the week of May 2, 2027)

CS 191 Schedule

Milestone	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Confirm details with the client												
Data collection and data cleaning (Use only partial dataset from the last 3 months)												
Design database structure												
Build database and backend API												
Build frontend												
Testing and polishing												
Software test report												

CS 192 Schedule

Milestone	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Set up full PhilGEPS dataset access															
First release (fully operational)															
Client meetings and evaluations															
Feature and accuracy improvements															
Testing and software test reports															
2nd release (revised from user comments)															
3rd release (improved from evaluation and maintenance)															

c. Project Metrics

To ensure the project is going well, the following metrics will be utilized:

Development and Process Metrics		
Metric	**Description**	**Target Value**
Schedule Variance	Actual milestone completion date - planned milestone completion date, measured in working days	*<= 0 days*
Requirements Volatility	Number of distinct requirements changed, added, or removed from the agreed baseline requirements	*< 15%*
Mean Time to Change	Average working days needed to analyze, fix, test, and deploy a routine correction. Exclude major feature changes.	*<= 5 days*
Defect Density	Number of distinct, verified defects discovered during system testing over total implemented function points	*< 0.15/FP*
Critical Defects	Unresolved critical defects that prevent core workflows or produce incorrect procurement data or price comparison for each phase	*0*

Operational, Usability, and Business Metrics		
Metric	**Description**	**Target Value**
Data Import Retention	Percentage of eligible source records successfully stored during import, excluding documented duplicates and intentional exceptions	*>= 99%*
Completed ICT Project Selection	Accuracy of including completed ICT projects and excluding non-ICT, incomplete, or unverified projects	*>=95%*
ICT Item Matching Quality	Percentage of reviewed search matches whose item descriptions are relevant to the requested ICT keyword	*>=90%*
Item Traceability	Percentage of reviewed ICT items linked to the correct original PhilGEPS notice containing their supporting description	*100%*
User Response Time	Time between a search, filter change, or map-pin selection and the appearance of usable results, excluding the initial map loading	*< 5 seconds user; < 8 seconds map*
User Task Success Rate	Percentage of users who independently complete each core task	*80%*

d. Effort Breakdown

The work will be divided among three developers. Each developer leads assigned components, while requirements review, integration, testing, and documentation remain shared responsibilities.

Module	Components	Developer	Responsibility
Requirements and System Design	*Requirements baseline, data availability assessment, database design, interface mockups, and acceptance criteria*	All	Confirm project scope, examine available PhilGEPS data, define eligibility and completion rules, and agree on the system structure and expected behavior
Data Acquisition and Preparation	*PhilGEPS data ingestion, record validation, duplicate handling, description normalization, and source references*	Ethan Kurt Espejo	Build the import process, preserve original record identifiers and source links, document exclusions, and prepare eligible records for storage
Data Storage, Location Mapping, and Search	*Database, agency location mapping, keyword search, geographic filters, and award-date filters*	Ambernie Salting	Store procurement records, associate agencies with supported administrative locations, and implement search and filtering
Procurement Analysis	*Grouping relevant ICT descriptions, project-level summaries, and assessment of available price information*	Ethan Kurt Espejo, Ambernie Salting	Implement analysis supported by the available data. Distinguish contract prices from item prices and document limitations affecting comparisons.
Interactive Map and Search Interface	*Map markers, zooming and panning, search controls, geographic filters, and agency purchase lists*	Jason Patrick Fangonil	Build the public map interface and connect its controls to the search services. Ensure displayed locations and purchase lists reflect the selected filters.
Purchase Details and Public Access	*Record details, source links, last-updated information, and responsive layouts*	Jason Patrick Fangonil	Present procurement information clearly, preserve traceability to original notices, label missing information, and supporting access without requiring an account

Integration, Testing, and Delivery	*Component integration, data checks, functional tests, performance checks, user testing, deployment, and documentation*	All	Verify the agreed metrics, resolve defects prepare the deployment, and complete technical documentation and user guidance

III. Project Risks

Take note of these legends for category and impact:

Category:
DQ – Data Quality and Availability
TR – Technical Reliability
PM – Project Management
UX – User Experience

Impact:
1 – Catastrophic
2 – Critical
3 – Marginal
4 – Negligible

Risks	Category	Probability	Impact	RMMM
Records may lack sufficient evidence that procurement projects are completed	DQ	*80%*	2	Define acceptable completion evidence and assess its availability across the dataset.
Missing contract amounts, award dates, and supplier details may prevent required displays, date filtering, and price analysis	DQ	*80%*	2	Profile missing fields before implementation and monitor usable-record coverage. Agree on features supported by the data
Budgets, quantities, and bundled items may be mistaken for actual item prices or comparable purchases	DQ	*70%*	3	Distinguish budgets from contract amounts, check units and invalid values, and exclude unsupported unit-price calculations or comparisons
Broad categories and ambiguous descriptions	DQ	*60%*	3	Apply documented classification rules, review

may cause incorrect ICT classification and or keyword matches				relevant and irrelevant examples, and refine rules or flag ambiguous matches.
Import, grouping, or filtering errors may lose records, double count projects, or produce inconsistent results	TR	*50%*	2	Validate parsing and record counts, distinguish items from projects, and repeat known result tests. Restore verified data when updates fail.
Large datasets or unavailable external services may slow or interrupt searches and map access	TR	*50%*	3	Monitor response times and service limits. Retain verified data and handle service failures clearly.
Technical difficulties, integration problems, or reduced developer availability may delay deliverable	PM	*60%*	2	Agree on interfaces, integrate early, and review progress weekly. Share knowledge, reassign blocked work, and reserve time for testing.
Users may struggle with core tasks or misunderstand incomplete and approximate information	UX	*40%*	3	Clearly label price meanings, dates, missing data, and location precision.

References

Philippine Government Electronic Procurement System. (n.d.). PhilGEPS. https://notices.philgeps.gov.ph/